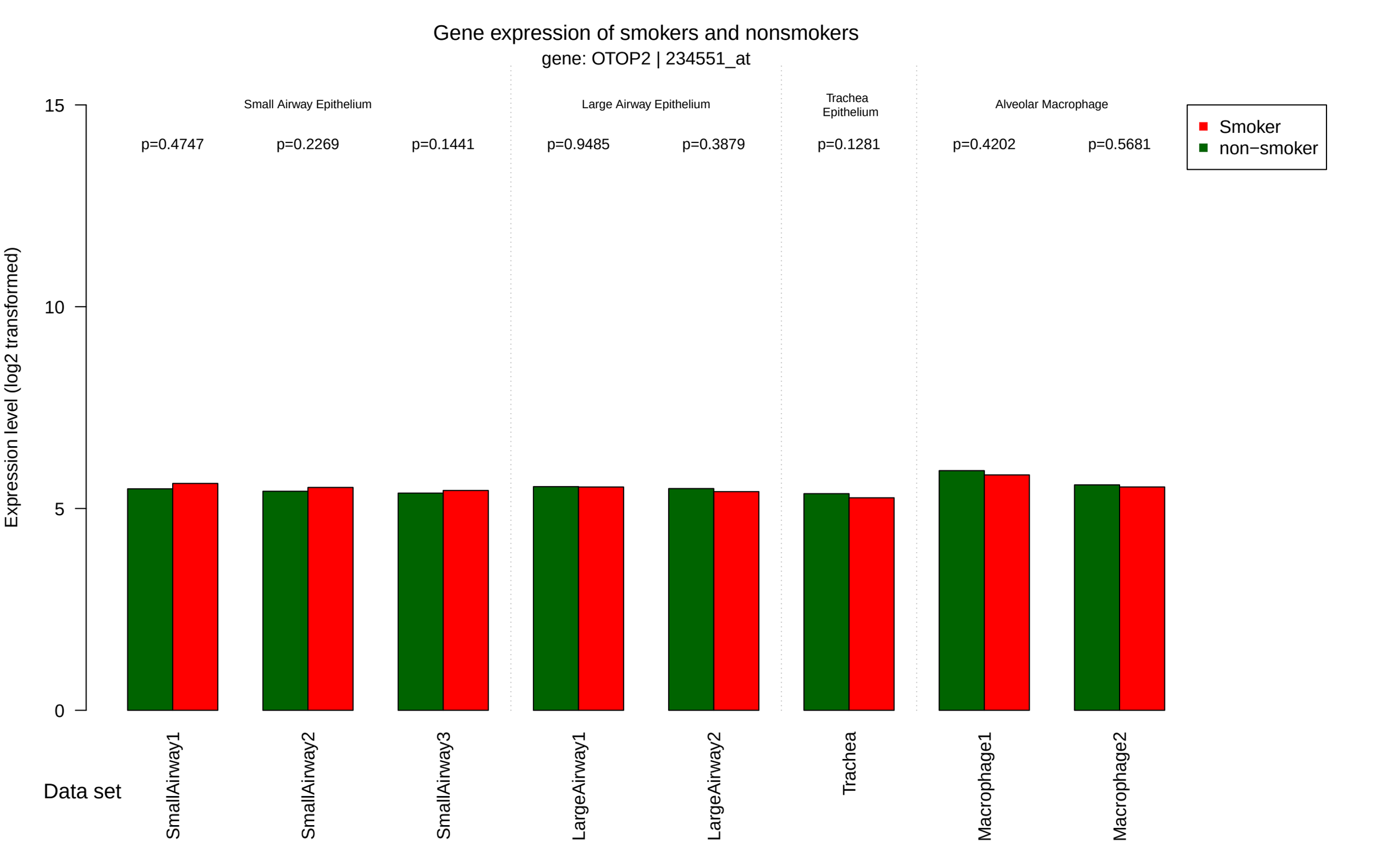
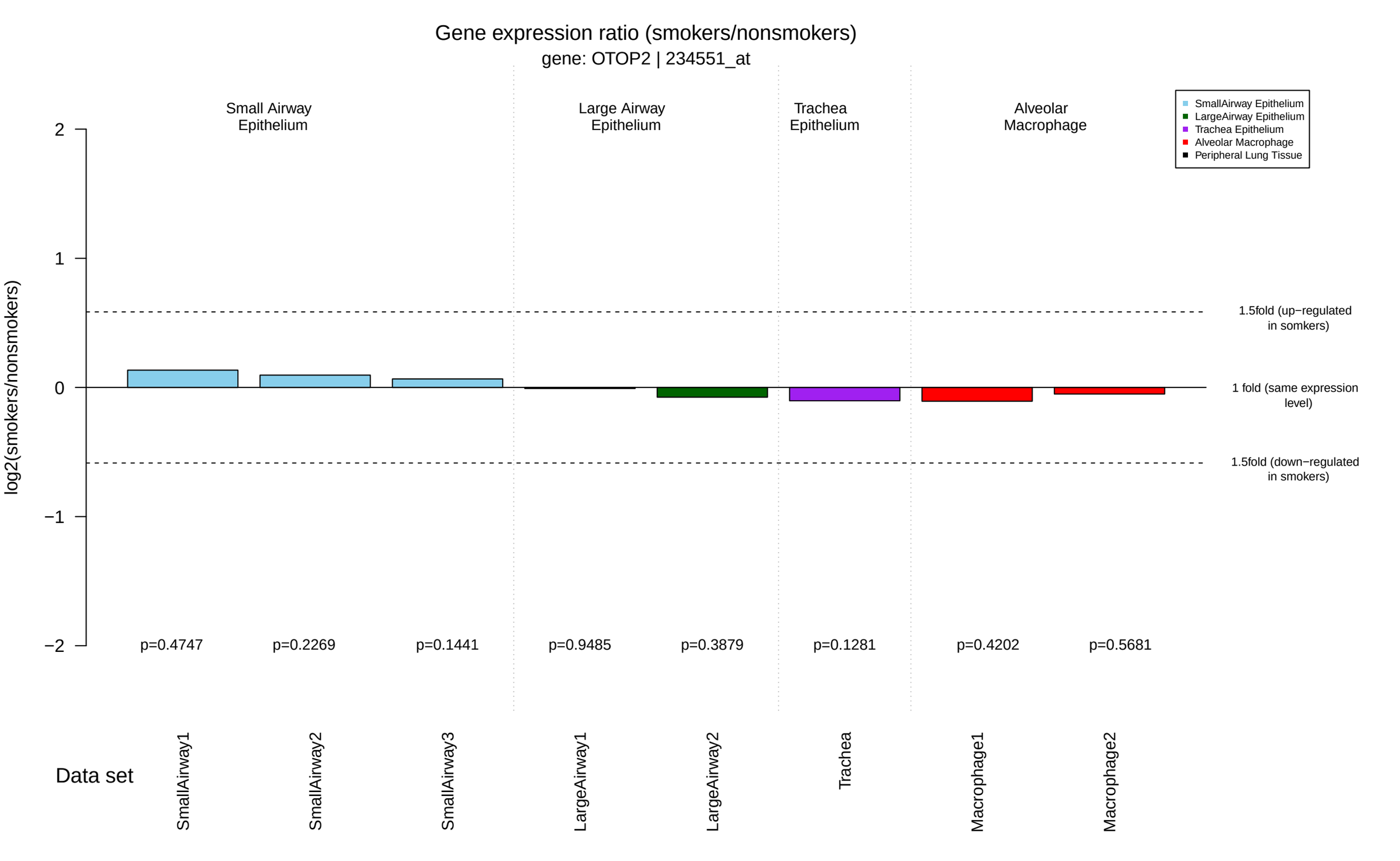
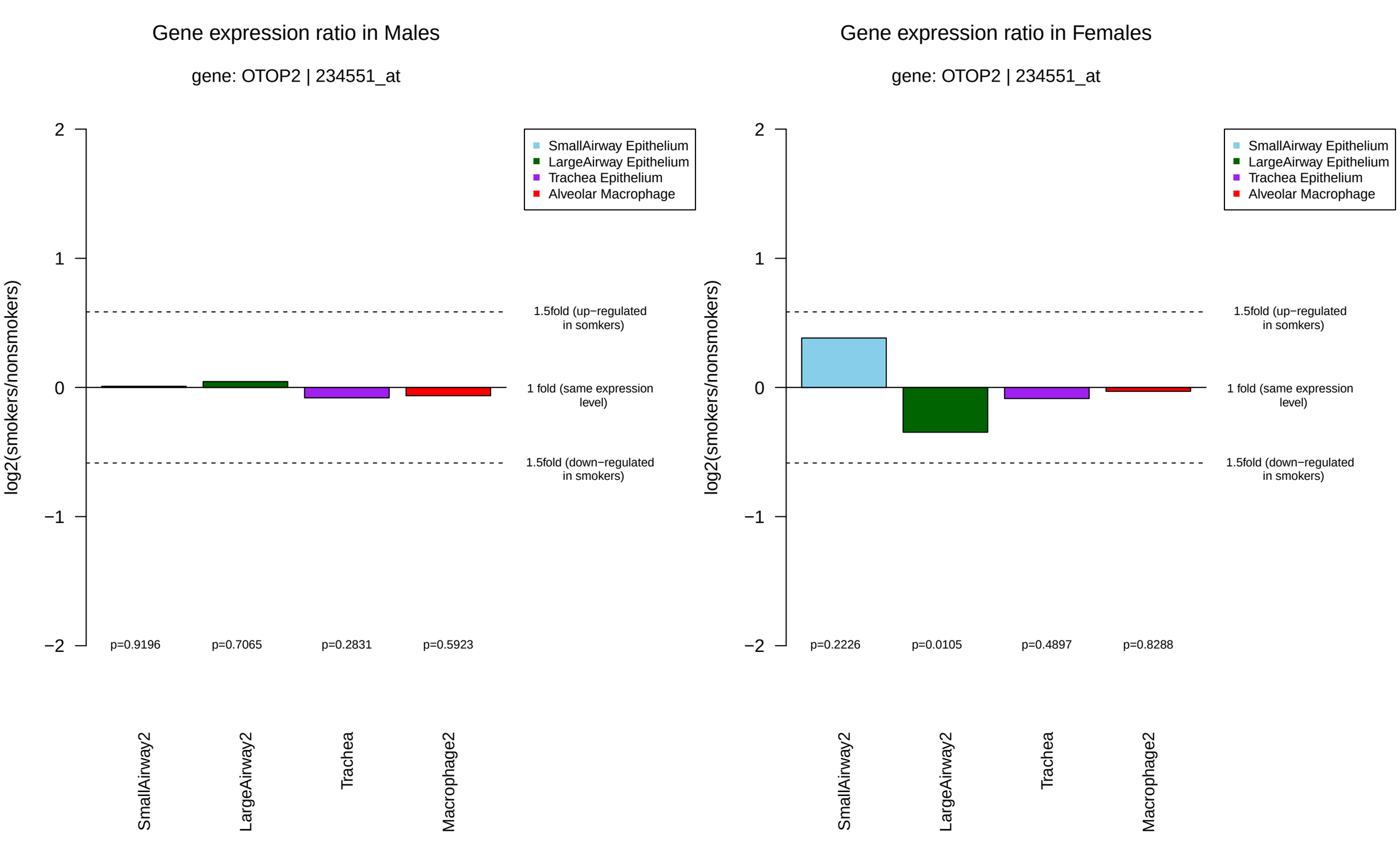
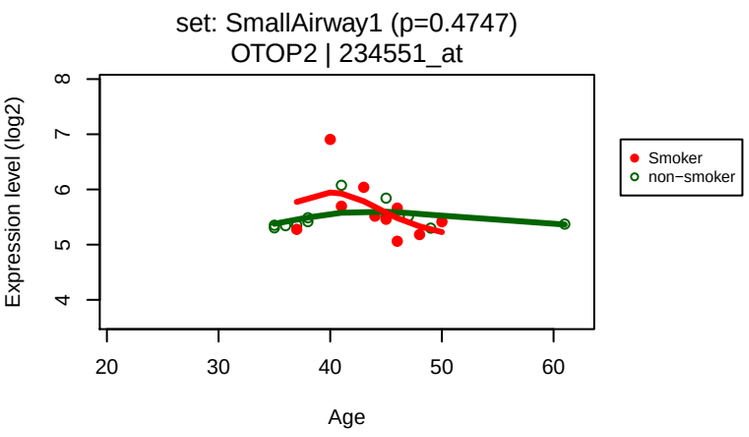
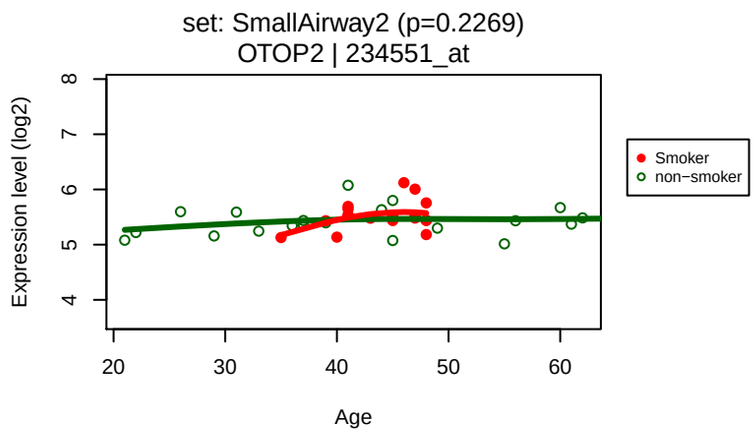
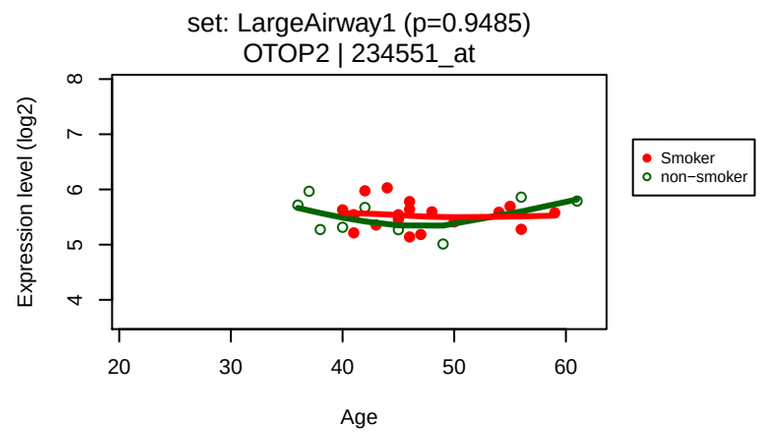

Gene expression of smokers and nonsmokers
gene: OTOP2 | 234551_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4747
p=0.2269
p=0.1441
p=0.9485
p=0.3879
p=0.1281
p=0.4202
p=0.5681
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

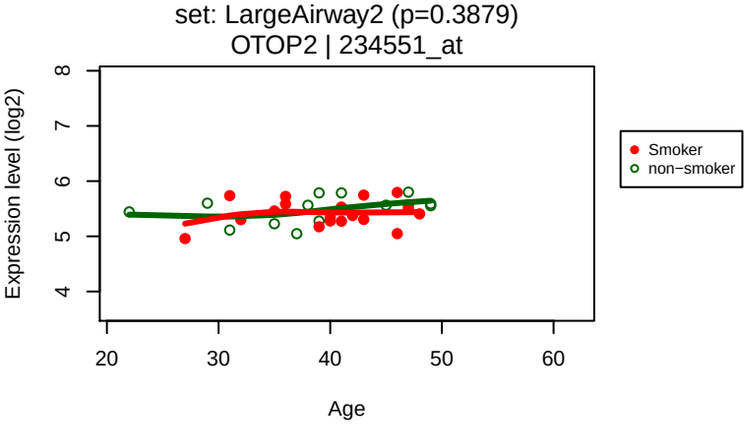

Gene expression ratio (smokers/nonsmokers)
gene: OTOP2 | 234551_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4747
p=0.2269
p=0.1441
p=0.9485
p=0.3879
p=0.1281
p=0.4202
p=0.5681
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

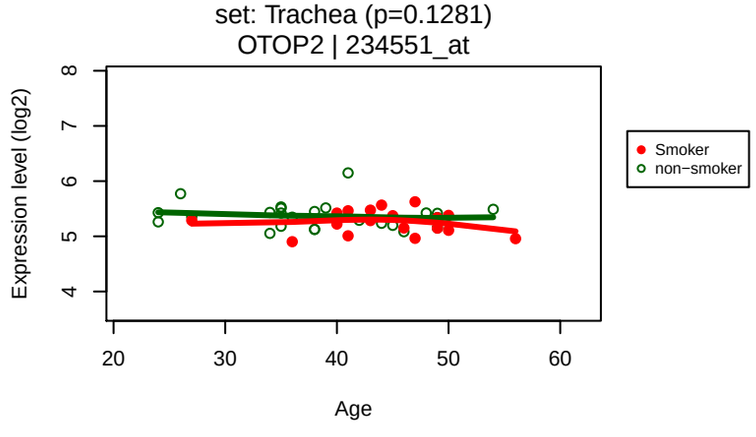

Gene expression ratio in Males
gene: OTOP2 | 234551_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9196
p=0.7065
p=0.2831
p=0.5923
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OTOP2 | 234551_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2226
p=0.0105
p=0.4897
p=0.8288
SmallAirway2
LargeAirway2
Trachea
Macrophage2

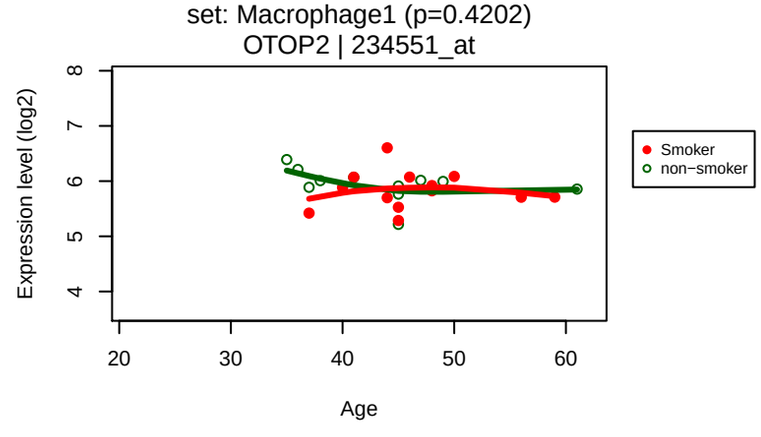

set: SmallAirway1 (p=0.4747)
OTOP2 | 234551_at
Expression level (log2)
Age
Smoker
non−smoker

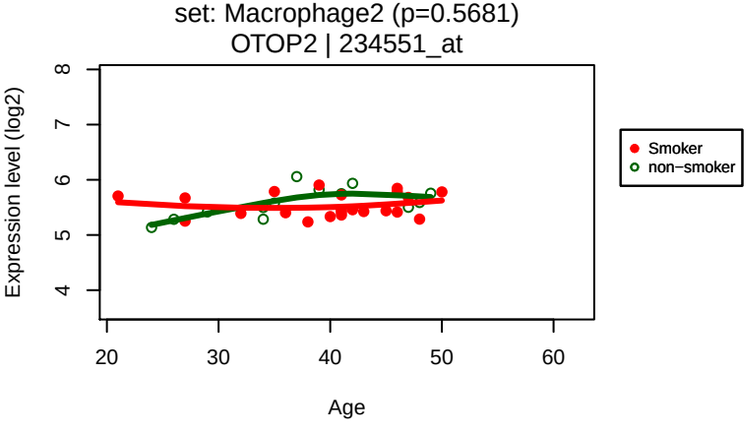

set: SmallAirway2 (p=0.2269)
OTOP2 | 234551_at
Expression level (log2)
Age
Smoker
non−smoker

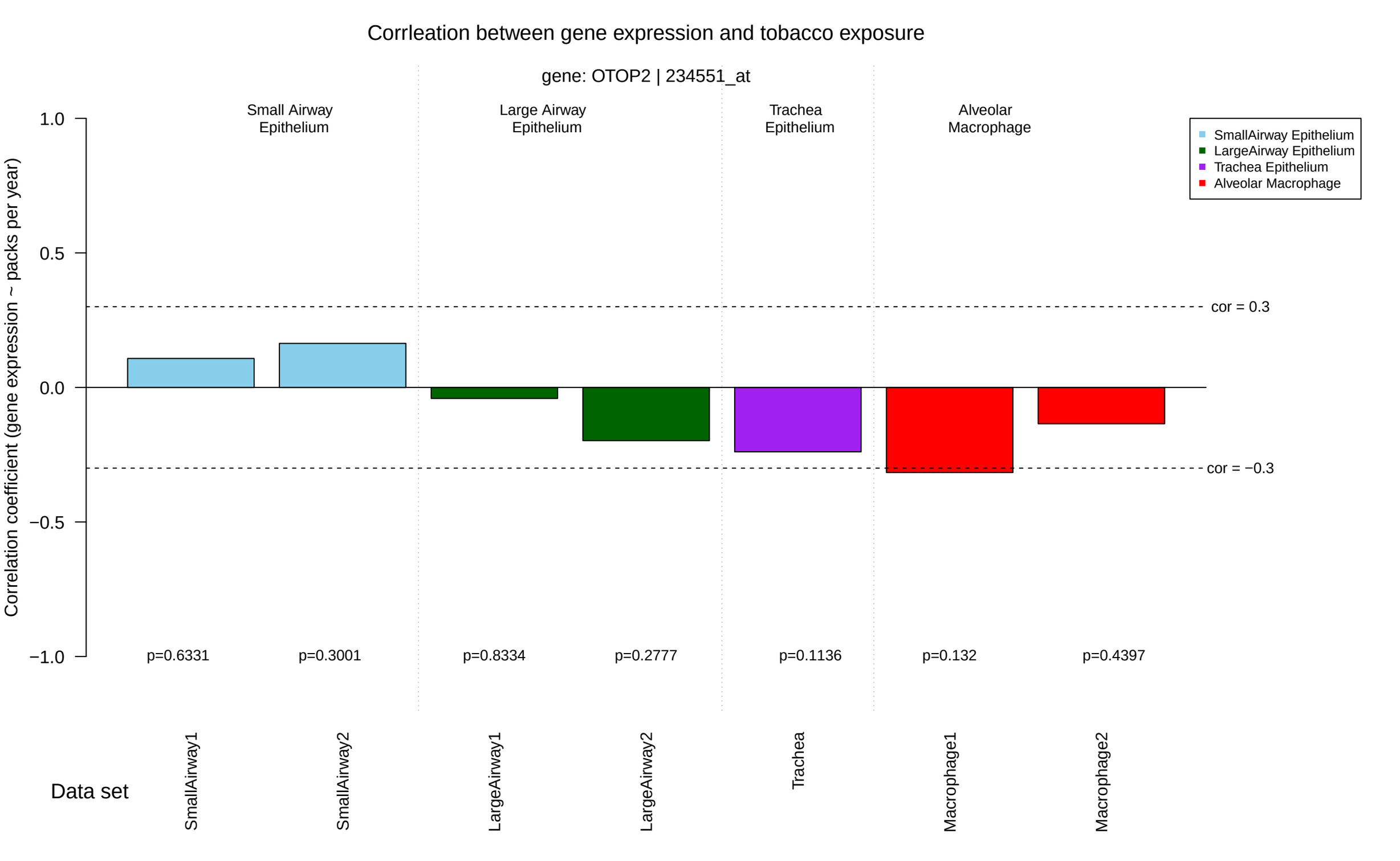

set: LargeAirway1 (p=0.9485)
OTOP2 | 234551_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3879)
OTOP2 | 234551_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1281)
OTOP2 | 234551_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4202)
OTOP2 | 234551_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5681)
OTOP2 | 234551_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OTOP2 | 234551_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6331
p=0.3001
p=0.8334
p=0.2777
p=0.1136
p=0.132
p=0.4397
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2